ME 351 Feedback Control Systems

Impulse Response of a Linear System

Dirk M. Luchtenburg | Fall 2026

The Unit Impulse

The unit impulse $\delta(t)$ is an idealized input concentrated at $t = 0$, with unit area:

$$\delta(t) = 0, \qquad t \neq 0,$$

and

$$\boxed{\int_{-\infty}^{\infty} \delta(t)\, dt = 1.}$$

It may be viewed as the limit of increasingly narrow pulses of unit area.

Its key property is the *sifting property*

$$\boxed{\int_{-\infty}^{\infty} x(\tau)\delta(t-\tau)\, d\tau = x(t).} \tag{1}$$

Therefore, any signal can formally be written as

$$\boxed{x(t) = \int_{-\infty}^{\infty} x(\tau)\delta(t-\tau)\, d\tau.} \tag{2}$$

This expression has an important interpretation:

> *Any input can be regarded as a continuous superposition of scaled and shifted impulses.*

Impulse Response

Consider a linear time-invariant (LTI) system. Its *impulse response* $h(t)$ is the output produced by a unit impulse with zero initial conditions:

$$\boxed{\delta(t) \quad \longrightarrow \quad h(t).}$$

Because the system is time invariant, shifting the impulse shifts the response:

$$\delta(t-\tau) \quad \longrightarrow \quad h(t-\tau).$$

Because the system is linear, scaling the impulse scales the response:

$$u(\tau)\delta(t-\tau) \quad \longrightarrow \quad u(\tau)h(t-\tau).$$

But the input itself can be written as

$$u(t) = \int_{-\infty}^{\infty} u(\tau)\delta(t-\tau)\,d\tau.$$

By superposition, the corresponding output must therefore be

$$\boxed{y(t) = \int_{-\infty}^{\infty} u(\tau)h(t-\tau)\,d\tau.} \tag{3}$$

This operation is called *convolution*:

$$\boxed{y(t) = (h * u)(t).} \tag{4}$$

For a causal system, the impulse response satisfies $h(t) = 0$ for $t < 0$. If the input is also zero for $t < 0$, the convolution integral simplifies to

$$\boxed{y(t) = \int_0^t h(t-\tau)u(\tau),d\tau, \qquad t \geq 0.}$$

Why the Impulse Response Matters

The impulse response completely characterizes an LTI system.

If $h(t)$ is known, the response to *any* input can be computed by convolution:

$$\boxed{u(t) \quad \xrightarrow{\text{LTI system}} \quad y(t) = h(t) * u(t).}$$

Thus, instead of solving the differential equation again for every new input, we determine the response to one special input—the impulse—and use superposition.

Connection to the Transfer Function

Taking the Laplace transform of the convolution equation,

$$Y(s) = H(s)U(s),$$

where

$$\boxed{H(s) = \mathcal{L}\{h(t)\}.}$$

Hence, with zero initial conditions,

$$\boxed{H(s) = \frac{Y(s)}{U(s)}.}$$

The transfer function and impulse response therefore contain the same input–output information:

$$\boxed{h(t) \quad \mathcal{L} \quad H(s).}$$

The impulse response describes the system in the *time domain*; the transfer function describes the same system in the *Laplace domain*.

State-Space Connection

For the state-space system

$$\dot{x} = Ax + Bu, \qquad y = Cx + Du,$$

with zero initial conditions,

$$y(t) = \int_0^t Ce^{A(t-\tau)}Bu(\tau)\,d\tau + Du(t).$$

Therefore, the impulse response is

$$\boxed{h(t) = Ce^{At}B + D\delta(t).} \tag{5}$$

For a strictly proper system ($D = 0$),

$$\boxed{h(t) = Ce^{At}B, \qquad t \geq 0.}$$

Thus the natural modes contained in e^{At} also appear directly in the impulse response.

Key Idea

$$\boxed{\begin{array}{c}\text{input} = \text{superposition of shifted impulses} \\ \Downarrow \\ \text{linearity} + \text{time invariance} \\ \Downarrow \\ y(t) = h(t) * x(t)\end{array}}$$

The impulse response is the fundamental time-domain description of an LTI system.